

Course 1 — Temperature Mo...

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✦ Course 1 — Temperature M...

Option 1

Option 2

Option 2 — Design from Block...

Block Diagram

✦ **Course 1 — Temperature Monitor Dev Board**

Course 1 is a hands-on hardware design exercise where you will design a complete **microcontroller-based temperature monitoring board with an OLED display**.

This is not a traditional lecture-style course. Instead of being told exactly what to do, you'll learn by **actively designing real hardware**, making decisions, and improving through iteration and feedback.

You can approach this course in two ways:

◆ **Option 1 — Follow-Along Design (EasyEDA, Same Board)**

In this option, you follow the **same overall project**, using the **same components and same design goals**, while watching a design walkthrough created in **EasyEDA**.

However, this is **not a slow, step-by-step tutorial**. The design process is shown at real working speed, sometimes slightly sped up, and the screen does not always show every detail at full resolution. That means you will still need to:

- Pause and rewind the video



This is intentional. The goal is not to copy clicks, but to **learn how design decisions are made and try to reproduce them yourself**.

This path is ideal if you:

- Want guidance but still want to think independently
- Are learning how real PCB workflows look
- Want to build the same board while developing problem-solving skills.

◆ Option 2 — Design from Specification (Any Tool)

In this option, you'll receive a **project specification and block diagram**, and you will design your own solution using **any design tool you prefer** — Altium, KiCad, EasyEDA, or something else.

You will:

- Choose your own components
- Design the power supply and protection strategy
- Decide how peripherals are connected
- Create your own PCB layout strategy

This path is ideal if you already have some experience and want to practice real engineering decision-making instead of following an existing design.

Goal of the Course

The goal is **not to build a perfect board**.

The goal is to:

- Start designing real hardware



- Improve through feedback and discussion

Real hardware design is iterative and imperfect, and this course is designed to reflect that reality.

You are encouraged to share your work in the community, review other designs, and participate in discussions. Some designs may also be selected for deeper review and walkthroughs, where we analyze what worked, what didn't, and how things could be improved.

This course is about building confidence, momentum, and practical skills — not just watching videos.